

Moneeb Hussain

+92 (321) 469-4262 • moneebhussain539@gmail.com • [LinkedIn](#) • [GitHub](#)

Achievements

- Winner of **Harvard CS50x** Puzzle Day as Team Lead, securing a perfect **10/10 score** across all puzzles. [🔗](#)
- Ranked in the **top 4%** of 10,000+ in HackerRank Orchestrate for a support-ticket triage agent. [🔗](#)
- Finished in the **top 8% worldwide** among 220+ teams in CodeSprint, hosted by **UCLA**. [🔗](#)
- **Pre-Finalist** and Special Honour recipient, International Computer Science Competition 2026. [🔗](#)
- Recognized as **Top Performer (Top 5%)** in the GenAI Program by Pak Angels USA & PEC Pakistan. [🔗](#)
- Winner of Line Following Robot and RoboWars in institutional robotics competitions. [🔗](#)
- Achieved **IELTS Academic Overall Band 7.0** (Listening: 7.5, Reading: 7.0, Writing: 6.5, Speaking: 6.5). [🔗](#)
- Competed in the **MIT Informatics Tournament** and **Calico** Coding Competition, hosted by UC Berkeley.
- Awarded a **merit-based laptop award** for academic excellence through a national youth initiative.

Education

University of Engineering and Technology, Lahore

2017 – 2021

BS in Mechatronics and Control Engineering

CGPA: 3.16/4.0

Selected Academic Tools: MATLAB/Simulink for feedback-control modeling and simulation; ROS for basic robot communication and simulation

Research & Academic Projects

Automatic Retail Checkout V-3 – AI-Based Retail Checkout System Senior-Year Thesis Project

[AI & Robotics Lab](#), UET Lahore | Advisor: [Muhammad Rzi Abbas](#) | [Vision-Based AI Automation](#)

- Developed a YOLOv4-tiny barcode-free retail checkout system for product detection, counting, and GUI billing.
- Integrated OpenCV inference, conveyor hardware, controlled lighting, and automated receipt generation.
- Built a custom conveyor-captured dataset of **3,500 images** across **70 retail-product classes**.
- Designed and built a **rotary-indexer mechanism** for automated, scalable image capture.
- Achieved **98.78%** detection accuracy in controlled testing, reducing checkout time from **~4s** to under **1s**.

Technologies/Hardware: Python, OpenCV, YOLOv4-tiny, Darknet, Arduino Uno, NEMA 17 Stepper Motor

HeatLens – Street-Level Urban Heat from Imagery FortyGuard Hackathon

Sep 2026

[Vision Pipeline & Model Design](#) | [Live App](#) | [Demo](#) | [Source Code](#)

- Mapped street photos to FortyGuard **2 m air ΔT** instead of satellite land-surface temperature.
- Used SegFormer fractions (canopy, asphalt, sky, buildings) to fit **°C-per-canopy** coefficients.
- Ranked streets for canopy investment by indicative ΔT drop if canopy rises toward 40%.

Technologies: Python, PyTorch, HuggingFace, SegFormer, FastAPI, Next.js, Mapillary API, FortyGuard API

Livestock AI – AI Livestock Health Assistant Generative AI Hackathon

May 2026

[Team Lead](#) | [Live App](#) | [System Design](#) | [Source Code](#)

- Led a multimodal AI triage system for early livestock-health guidance in under-resourced farming communities.
- Designed workflows for symptom chat, image observations, voice input, case history, and outbreak reporting.
- Implemented structured AI outputs with severity labels, care steps, safety validation, and vet escalation.

Technologies: Next.js, FastAPI, Supabase, Groq, Gemini Vision, React Speech Recognition, OpenStreetMap

Professional Experience

Septem Systems Lahore, Pakistan

Oct 2023 – Present

Software Engineer Level II

- Built production workflows across front-end, backend APIs, secure workflows, CI/CD, and GCP deployments.
- Reduced **API latency from 3.9s to 1.2s** through SQL optimization, legacy refactoring, and debugging.
- Recognized as **Employee of the Quarter** for problem-solving and reliable delivery. 🔗

Style Textile Pvt. Ltd. Lahore, Pakistan

Aug 2021 – Aug 2023

Engineer

- Analyzed PLC logic, electrical schematics, and I/O behavior to troubleshoot automation faults.
- Validated hardware-control integration during ASRS warehouse automation commissioning and deployment.

Teaching & Mentorship

iCodeGuru Remote – Silicon Valley, USA

Feb 2026 – Present

Technical Trainer & Moderator 🔗

- Mentored **80+ underprivileged learners** across programming, data structures, and AI-focused training.
- Co-designed and delivered a 6-week **Python-to-GenAI** curriculum for a cohort of **150+** enrolled learners.
- Taught Python, Operating Systems, AI, and Machine Learning through live bootcamp sessions.
- Designed and delivered a two-part **MATLAB/Simulink** control systems workshop series, progressing from **PID** fundamentals to closed-loop robustness against disturbance, sensor noise, and measurement delay.

Technical Skills

AI, ML & GenAI: Generative AI, Multimodal AI, LLM Agents, Multi-Agent Systems

Computer Vision: OpenCV, Object Detection, Image Processing, Image Annotation, Data Preprocessing

Programming & Scientific Computing: Python, MATLAB, JavaScript, TypeScript, Pandas, NumPy, Matplotlib

Robotics, Control & Automation: ROS, Simulink, Arduino, Control Systems, Sensor/Actuator Integration

Backend & Cloud Systems: FastAPI, Node.js, NestJS, Next.js, React.js, REST APIs, SQL, Docker, GCP, CI/CD

Databases & Tools: MySQL, MongoDB, PostgreSQL, Supabase, Git, GitHub, GitLab

Leadership & Community Service

HSE Medical Camps Community Health Volunteer

Aug 2022

16 Hours of Community Service 🔗

- Coordinated logistics, patient flow, supplies, and OPD support for medical camps serving **200–300 patients**.

AMAL Academy Career-Prep Fellow

Oct 2019 – Dec 2019

Leadership Fellowship 🔗

- Graduated with Honors from a **Stanford-supported** leadership fellowship and led a peer social-impact project.